

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250285491

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

MORISHITA; Haruka et al.

ACCOMMODATION SYSTEM, CONTROL METHOD FOR ACCOMMODATION SYSTEM, AND NON-TRANSITORY TANGIBLE STORAGE MEDIUM

Abstract

An accommodation system includes: a plurality of accommodation boxes each having a door and disposed so as to be arranged in an in-plane direction; a display device configured to display an image on a door of an accommodation box of the plurality of accommodation boxes; and a controller configured to acquire information on the accommodation box for which an operation of opening the door has been performed, set the door, on which the image is to be displayed, in accordance with a position of the accommodation box identified by the acquired information, and control the display device to display the image on the door.

Inventors: MORISHITA; Haruka (Himeji-shi, JP), AOKI; Soshi (Himeji-shi, JP)

Applicant: Glory Ltd. (Himeji-shi, JP)

Family ID: 94970173

Assignee: Glory Ltd. (Himeji-shi, JP)

Appl. No.: 19/074487

Filed: March 10, 2025

Foreign Application Priority Data

JP	2024-037606	Mar. 11, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: G07F17/12 (20060101); G07C9/00 (20200101); G07F17/16 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority under 35 U.S.C. Section 119 of Japanese Patent Application No. 2024-037606 filed Mar. 11, 2024, entitled “ACCOMMODATION SYSTEM, CONTROL METHOD FOR ACCOMMODATION SYSTEM, and PROGRAM”. The disclosure of the above application is incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention relates to an accommodation system having a plurality of accommodation boxes disposed so as to be arranged in an in-plane direction, a control method for the accommodation system, and a non-transitory tangible storage medium storing computer executable instruction for causing a controller of the accommodation system to execute predetermined functions.

Description of Related Art

[0003] In stations and commercial facilities, lockers for temporary deposit of luggage are installed. Usually, in a locker, a plurality of accommodation boxes are disposed so as to be arranged in a matrix shape in an in-plane direction. Recently, the empty status and unlocking or locking of each accommodation box are managed in a centralized manner by a control assembly. A user can use a desired accommodation box via an interface connected to the control assembly.

[0004] Japanese Laid-Open Patent Publication No. 2023-137333 below describes an accommodation apparatus that includes: a locker in which a display is installed at a door of each accommodation box; and a control apparatus that controls an open/close state of the locker. In this accommodation apparatus, an independent display mode in which an image is independently displayed for each door, and a whole-screen display mode in which an integrated image is displayed in coordination over the entirety of a plurality of displays are executed by the control apparatus. In the independent display mode, usage information (information on empty/in-use) for using the locker is provided, and in the whole-screen display mode, a predetermined advertisement is displayed.

[0005] For an advertiser, it is desirable that the advertisement displayed on the locker is viewed by as many passersby as possible. However, many passersby are highly likely to pass by the locker without being aware thereof. In addition, when the passage in front of the locker is crowded, the advertisement itself need not necessarily be seen by the passersby in the first place.

[0006] Even in such a circumstance, a user who uses the locker is highly likely to pay attention to the advertisement. In this case, as in Japanese Laid-Open Patent Publication No. 2023-137333 above, if the advertisement is displayed in a large size targeting passersby, the distance between the user and the advertisement is too close for the size of the advertisement, and thus, the user cannot smoothly understand the content of the advertisement. In addition, for the user, if the advertisement is displayed at a position away from the accommodation box that the user himself/herself uses, the user cannot naturally refer to the advertisement when using the accommodation box. Therefore, the advertisement needs to be displayed on the door of each accommodation box such that the user can more naturally refer to the same when using the accommodation box.

SUMMARY OF THE INVENTION

[0007] A first aspect of the present invention relates to an accommodation system. The accommodation system according to this aspect includes: a plurality of accommodation boxes each having a door and disposed so as to be arranged in an in-plane direction; a display device

configured to display an image on a door of an accommodation box of the plurality of accommodation boxes; and a controller configured to acquire information on the accommodation box for which an operation of opening the door has been performed, set the door, on which the image is to be displayed, in accordance with a position of the accommodation box identified by the acquired information, and control the display device to display the image on the door.

[0008] In the accommodation system according to the present aspect, in accordance with the position of the accommodation box for which the operation regarding opening of the door has been performed by the user, the door for displaying the image is set. Therefore, the user can naturally refer to the image such as an advertisement at the time of his/her use of the accommodation box.

[0009] A second aspect of the present invention relates to an accommodation system including a plurality of accommodation boxes each having a door and disposed so as to be arranged in an in-plane direction, and a display device configured to display an image on a door of an accommodation box of the plurality of accommodation boxes. The control method according to this aspect includes: acquiring information on the accommodation box for which an operation of opening the door has been performed; setting the door, on which the image is to be displayed, in accordance with a position of the accommodation box identified by the information; and controlling the display device to display the image on the set door.

[0010] A third aspect of the present invention relates to a non-transitory tangible storage medium storing computer executable instructions for causing a controller of an accommodation system including a plurality of accommodation boxes each having a door and disposed so as to be arranged in an in-plane direction, and a display device configured to display an image on a door of an accommodation box of the plurality of accommodation boxes, to execute a process comprising: acquiring information on the accommodation box for which an operation of opening the door thereof has been performed; setting the door on which the image is to be displayed, in accordance with a position of the accommodation box identified by the information; and controlling the display device to display the image on the set door.

[0011] The effects and the significance of the present invention will be further clarified by the description of the embodiments below. However, the embodiments below are merely examples for implementing the present invention. The present invention is not limited to the description of the embodiments below in any way.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1 shows a configuration of an accommodation system according to Embodiment 1;

[0013] FIG. 2 is a front view showing a configuration of a locker apparatus according to Embodiment 1;

[0014] FIG. 3A and FIG. 3B are each a front view showing a configuration of a panel according to Embodiment 1;

[0015] FIG. 4 is a front view showing a configuration of a settlement assembly according to Embodiment 1;

[0016] FIG. 5 is a block diagram showing a configuration of a circuitry of the accommodation system according to Embodiment 1;

[0017] FIG. 6 shows a configuration of an initial screen displayed on an interface of the settlement assembly according to Embodiment 1;

[0018] FIG. 7 shows a configuration of a confirmation screen displayed on the interface of the settlement assembly according to Embodiment 1;

[0019] FIG. 8 shows a configuration of a settlement screen displayed on the interface of the settlement assembly according to Embodiment 1;

[0020] FIG. **9** shows a configuration of a take-out screen displayed on the interface of the settlement assembly according to Embodiment 1;

[0021] FIG. **10A** is a flowchart showing a display process of an advertisement image according to Embodiment 1;

[0022] FIG. **10B** is a flowchart showing a process of a second mode according to Embodiment 1;

[0023] FIG. **11A** shows an example of a setting method of doors to be used for displaying the advertisement image in the second mode according to Embodiment 1;

[0024] FIG. **11B** shows a setting example of a locker number for each accommodation box according to Embodiment 1;

[0025] FIG. **12A** to FIG. **12C** each schematically show display methods of the advertisement image in a first mode and the second mode according to Embodiment 1;

[0026] FIG. **13** shows a setting example of a display range in the first mode according to Embodiment 1;

[0027] FIG. **14** shows a setting example of a display range in the second mode according to Embodiment 1;

[0028] FIG. **15** shows a setting example of a display range in the second mode according to Embodiment 1;

[0029] FIG. **16** shows a setting example of a display range in the second mode according to Embodiment 1;

[0030] FIG. **17** shows a setting example of a display range in the second mode according to Embodiment 1;

[0031] FIG. **18** shows a setting method of a display range according to Modification 1 of Embodiment 1;

[0032] FIG. **19** shows a setting method of a display range according to Modification 1 of Embodiment 1;

[0033] FIG. **20** shows a setting method of a display range according to Modification 1 of Embodiment 1;

[0034] FIG. **21** is a flowchart showing a process of the second mode according to Modification 2 of Embodiment 1;

[0035] FIG. **22** shows a range (display range) of doors re-set in the process of the second mode according to Modification 2 of Embodiment 1;

[0036] FIG. **23A** is a flowchart showing a display process of the advertisement image according to Modification 3 of Embodiment 1;

[0037] FIG. **23B** is a flowchart showing a process of the second mode according to Modification 3 of Embodiment 1;

[0038] FIG. **24** is a flowchart showing a display process of the advertisement image according to Modification 4 of Embodiment 1;

[0039] FIG. **25** shows a configuration of the accommodation system according to Embodiment 2; and

[0040] FIG. **26** shows a configuration of the accommodation system according to another configuration example.

DETAILED DESCRIPTION

[0041] Hereinafter, embodiments of the present invention will be described with reference to the drawings.

Embodiment 1

[0042] FIG. **1** shows a configuration of an accommodation system **1** according to Embodiment 1.

[0043] The accommodation system **1** includes a locker apparatus **10** and a control assembly **20**. The locker apparatus **10** and the control assembly **20** are installed in a station, a commercial facility, and the like.

[0044] The locker apparatus **10** includes a plurality of accommodation boxes **11** arranged in a

matrix shape in an in-plane direction, and a settlement assembly **12**. Each accommodation box **11** has a door **11a** that can be opened and closed. Here, each door **11a** has a left-opening configuration in which the door **11a** rotates about the vicinity of the left side thereof. When the door **11a** is opened, the accommodation space for luggage is opened. Each door **11a** can be locked. When luggage is accommodated in the accommodation space and then settlement is performed by using the settlement assembly **12**, the door **11a** of the accommodation space is locked. The settlement assembly **12** includes an interface **121** described later.

[0045] The control assembly **20** is implemented by an information processing apparatus such as a personal computer and is built in the locker apparatus **10**. However, not limited thereto, the control assembly **20** may be mounted, as a control board, to the settlement assembly **12**. The control assembly **20** controls the settlement assembly **12** to perform settlement according to use of the locker apparatus **10**. When accommodation and settlement of luggage with respect to a certain accommodation box **11** has been performed, the control assembly **20** locks the door **11a** of the accommodation box **11**.

[0046] The control assembly **20** performs communication with a server **30** via an external communication network **40** such as the Internet. The control assembly **20** acquires image information from the server **30**. In the present embodiment, this image information is an advertisement image including advertisement information. The advertisement image is provided from an advertiser to a manager of the server **30**, and the manager of the server **30** registers the advertisement image into the server **30**. The control assembly **20** stores the image information of the advertisement image acquired from the server **30** and causes the advertisement image to be displayed on a predetermined door **11a** of the locker apparatus **10**, by using the stored image information.

[0047] FIG. **2** is a front view showing a configuration of the locker apparatus **10**.

[0048] A display **13** is installed at the surface of each door **11a**. The display **13** is implemented by a liquid crystal panel, for example. The size of the display **13** is slightly smaller than the size of the surface of the door **11a**. The control assembly **20** in FIG. **1** causes the display **13** of a predetermined door **11a** to display the above-described advertisement image acquired from the server **30**. At the right end of each door **11a**, a handle **11b** that is gripped when the door **11a** is opened and closed, is installed. At the upper right corner of each door **11a**, a label (not shown) indicating a locker number is affixed. In a space on the right side of each door **11a**, a panel **14** that is used when the door **11a** is used, is disposed.

[0049] FIGS. **3A**, **3B** are each a front view showing a configuration of the panel **14**.

[0050] FIG. **3A** shows a state of the panel **14** when the accommodation box **11** immediately to the left of the panel **14** is in an empty state. FIG. **3B** shows a state of the panel **14** when the accommodation box **11** immediately to the left of the panel **14** is in an in-use state.

[0051] The panel **14** includes a lighting region **141**, a message region **142**, a temporary lock operation unit **143**, and a keyhole **144**. The keyhole **144** is used by the manager of the locker apparatus **10** to open the door **11a** with a key in an emergency.

[0052] The lighting region **141** is lit up when the accommodation box **11** immediately to the left of the panel **14** is in an empty state, to inform the user that the accommodation box **11** can be used. Here, the lighting region **141** is lit up in green. Accordingly, as shown in FIG. **3A**, the lighting region **141** emits light in green, and characters “usable” in the lighting region **141** appear. When the accommodation box **11** immediately to the left of the panel **14** is in use, the lighting region **141** turns off and the characters “usable” disappear as shown in FIG. **3B**. The user can understand a usable accommodation box **11** by referring to the lighting region **141**.

[0053] In the message region **142**, an operation procedure for the accommodation box **11** and the panel **14** are shown. That is, the message region **142** indicates that luggage is to be put in the accommodation box **11** immediately to the left of the panel **14**, and a lever **143a** of the temporary lock operation unit **143** is to be lowered.

[0054] The temporary lock operation unit **143** has the lever **143a** for a temporary lock operation, and a mark **143b** indicating an unlocked state. The lever **143a** and the mark **143b** are integrally movable in the up-down direction along the quadrangular-shaped opening forming the contour of the temporary lock operation unit **143**.

[0055] When the user has placed the luggage in an accommodation box **11** at the time of deposit, the user downwardly pushes the lever **143a** from the state in FIG. 3A to the state in FIG. 3B. Accordingly, the door **11a** of the accommodation box **11** is temporarily locked. This operation is detected by a sensor of a locking device **212** (see FIG. 5), and the detection result is outputted to the control assembly **20**. In response to this, the control assembly **20** causes the lighting region **141** of the panel **14** to flicker as described later.

[0056] FIG. 4 is a front view showing a configuration of the settlement assembly **12**.

[0057] The settlement assembly **12** includes the interface **121**, a barcode reader **122**, a card reader **123**, a banknote inlet/outlet **124**, a coin inlet **125**, a return lever **126**, a coin outlet **127**, a first ticket issuing port **128a**, and a second ticket issuing port **128b**.

[0058] The interface **121** displays predetermined information and receives an input with respect to the display. The interface **121** is implemented by a touch panel, for example. The barcode reader **122** reads a barcode on a take-out ticket issued from the second ticket issuing port **128b** at the time of cash settlement. Here, a two-dimensional barcode is printed on the take-out ticket. In this case, the barcode reader **122** is a two-dimensional barcode reader. However, not limited thereto, the barcode printed on the take-out ticket may be a one-dimensional barcode.

[0059] The card reader **123** reads information from an IC card. Here, the card reader **123** reads user information from an IC card of the user, by an RFID (Radio Frequency Identification) method. Therefore, the IC card of the user has a configuration according to the RFID method. Not limited thereto, the card reader **123** may be configured to read user information from a card in which user information is magnetically recorded.

[0060] The banknote inlet/outlet **124** is used when the user deposits banknotes and banknotes of change or unfit notes are dispensed during a settlement process. The coin inlet **125** is used when the user deposits coins during a settlement process. The return lever **126** is used when the user causes coins that have been put in the coin inlet **125** to be returned. The coin outlet **127** is used when coins of change or deformed coins are dispensed during a settlement process.

[0061] From the first ticket issuing port **128a**, a use certificate is issued when the user has performed settlement by using an IC card at the time of deposit of luggage. From the second ticket issuing port **128b**, a take-out ticket is issued when the user has performed settlement by cash at the time of deposit of luggage. The take-out ticket is used when the deposited luggage is taken out from the accommodation box **11**. In this case, the barcode printed on the take-out ticket serves as the key for unlocking the door **11a**. When settlement has been performed by using an IC card, unlocking is performed by using the IC card. That is, in this case, the IC card serves as the key for unlocking the door **11a**.

[0062] FIG. 5 is a block diagram showing a configuration of a circuitry of the accommodation system **1**.

[0063] The accommodation system **1** includes a controller **201**, a storage **202**, and a communication module **203**, as components of the control assembly **20**.

[0064] The controller **201** includes an arithmetic processing circuit such as a CPU (Central Processing Unit), and controls each display **13** of the locker apparatus **10**, according to a program stored in the storage **202**. The storage **202** includes memories such as a ROM (Read Only Memory), a RAM (Random Access Memory), and a hard disk, and stores therein the above program. The storage **202** is used as a work region when the controller **201** executes the above program.

[0065] The communication module **203** transmits image information of an advertisement image to a transmitter **230** through control from the controller **201**. In addition, the communication module

203 performs communication with a controller **221** of the settlement assembly **12** through control from the controller **201**. Further, the communication module **203** performs communication with the server **30** via the external communication network **40** shown in FIG. **1**.

[0066] The storage **202** stores therein layout information indicating the layout of the accommodation boxes **11** in the locker apparatus **10**, and locker management information in which each accommodation box **11** on the layout and a locker number are associated with each other. The setting of these pieces of information is performed by a manager or an installer of the locker apparatus **10**. The setting may be performed by connecting a dedicated terminal to the control assembly **20**, or may be performed by a method in which a setting screen is displayed on the interface **121**.

[0067] The locker apparatus **10** includes a set of a display **13**, a light emission device **211**, the locking device **212**, and an open/close sensor **213**, for each accommodation box **11**. That is, the locker apparatus **10** includes the sets of the display **13**, the light emission device **211**, the locking device **212**, and the open/close sensor **213** by the number of the accommodation boxes **11** disposed in the locker apparatus **10**.

[0068] The light emission device **211** includes a light emission source for the lighting region **141** shown in FIGS. **3A**, **3B**, and a control circuit therefor. The light emission source for the lighting region **141** is implemented by an LED that emits green light, for example.

[0069] The locking device **212** includes a lock mechanism that locks the door **11a** of a corresponding accommodation box **11** and a control circuit therefor, and performs locking and unlocking with respect to the door **11a** through control from the controller **221** of the settlement assembly **12**. The temporary lock operation unit **143** shown in FIGS. **3A**, **3B** is included in the lock mechanism. As described above, the locking device **212** includes a sensor that detects an operation performed on the lever **143a**, and transmits the detection result by the sensor to the controller **221** of the settlement assembly **12**.

[0070] The open/close sensor **213** detects opening/closing of the door **11a** of a corresponding accommodation box **11**, and outputs the detection result to the controller **221** of the settlement assembly **12**.

[0071] The settlement assembly **12** includes the controller **221**, a storage **222**, a communication module **223**, a depositing and dispensing machine **224**, and a ticket issuing device **225**, in addition to the interface **121**, the barcode reader **122**, and the card reader **123** shown in FIG. **4**.

[0072] The controller **221** includes an arithmetic processing circuit such as a CPU, and controls each component according to a program stored in the storage **222**. The storage **222** includes memories such as a ROM and a RAM, and stores therein the above program. The storage **222** is used as a work region when the controller **221** executes the above program. The storage **222** stores therein layout information and locker management information similar to the above.

[0073] The communication module **223** performs communication with the controller **201** of the control assembly **20** through control from the controller **221**. In addition, the communication module **223** performs communication with the light emission device **211**, the locking device **212**, and the open/close sensor **213** of each accommodation box **11** through control from the controller **221**.

[0074] The depositing and dispensing machine **224** performs depositing and dispensing of banknotes and coins with respect to the banknote inlet/outlet **124**, the coin inlet **125**, and the coin outlet **127**. The depositing and dispensing machine **224** has a transport mechanism that transports banknotes and coins along transport paths, a recognition unit that recognizes denominations of banknotes and coins on the transport paths, and a plurality of banknote accommodation boxes and a plurality of coin accommodation boxes that accommodate deposited banknotes and coins for each denomination. The transport path for banknotes is connected to the banknote inlet/outlet **124** in FIG. **4** and the plurality of banknote accommodation boxes. The transport path for coins is connected to the coin inlet **125** and the coin outlet **127** in FIG. **4** and the plurality of coin

accommodation boxes.

[0075] The ticket issuing device **225** issues the use certificate and the take-out ticket described above through the first ticket issuing port **128a** and the second ticket issuing port **128b** in FIG. **4**. The ticket issuing device **225** includes: a printing unit that prints a two-dimensional barcode to serve as key information for unlocking, a locker number, and use details on a roll of paper; and a cutting unit that cuts the roll of paper on which printing has been performed by the printing unit into a predetermined length, to generate a corresponding use certificate and a corresponding take-out ticket.

[0076] The transmitter **230** outputs image information of an advertisement image received from the controller **201** of the control assembly **20**, to a corresponding display **13**. The controller **201** transmits, to the transmitter **230**, image information to be displayed by each of the displays **13** serving as the display target, together with the identification information of the display **13**. The transmitter **230** outputs the image information to a display **13** corresponding to the received identification information. Accordingly, an image forming the advertisement image is displayed on the corresponding display **13**. In the transmitter **230** and the control assembly **20**, setting for performing of such image information distribution has been performed in advance by the installer of the accommodation system **1**.

[0077] A relay machine **240** outputs a control signal received from the controller **221** of the settlement assembly **12** to a corresponding light emission device **211** or a corresponding locking device **212**, and outputs detection signals received from the locking device **212** and the open/close sensor **213** to the controller **221**.

[0078] When a predetermined light emission device **211** is caused to light up or flicker, the controller **221** outputs a control signal therefor, together with the identification information of the corresponding light emission device **211**, to the relay machine **240**. The relay machine **240** outputs the control signal to the light emission device **211** corresponding to the received identification information. Accordingly, the corresponding light emission device **211** performs an operation of lighting up or flickering.

[0079] When a certain locking device **212** is to be locked, the controller **221** outputs a control signal therefor, together with the identification information of the corresponding locking device **212**, to the relay machine **240**. The relay machine **240** outputs the control signal to the locking device **212** corresponding to the received identification information. Accordingly, the corresponding locking device **212** performs an operation of locking.

[0080] When a certain locking device **212** has been temporarily locked, the locking device **212** outputs a detection signal thereof, to the relay machine **240**. The relay machine **240** transmits the received detection signal, together with the identification information of the locking device **212**, to the controller **221**. Accordingly, the controller **221** recognizes that the door **11a** corresponding to the locking device **212** has been temporarily locked.

[0081] When a certain door **11a** has been opened or closed, the open/close sensor **213** of the door **11a** outputs a detection signal thereof to the relay machine **240**. The relay machine **240** transmits the received detection signal, together with the identification information of the open/close sensor **213**, to the controller **221**. Accordingly, the controller **221** recognizes that the door **11a** corresponding to the open/close sensor **213** has been opened or closed.

[0082] In the relay machine **240** and the settlement assembly **12**, setting for performing such signal transmission/reception and recognition has been performed in advance by the installer of the accommodation system **1**.

<Deposit of Luggage>

[0083] A flow of deposit of luggage to the locker apparatus **10** will be described.

[0084] FIG. **6** shows a configuration of an initial screen **300** displayed on the interface **121** of the settlement assembly **12**.

[0085] The initial screen **300** has a message **301** that urges selection of a use item (deposit/take

out/balance inquiry/method for using locker), a selection button **302** for selecting deposit of luggage, a selection button **303** for selecting take-out of luggage, a selection button **304** for performing balance inquiry of an IC card, and a selection button **305** for referring to how to use the locker. When depositing luggage to the locker apparatus **10**, the user touches the selection button **302**. This makes a shift to a sequence for depositing luggage.

[0086] When the user wants to confirm the balance of the amount of money charged in his/her IC card before depositing luggage, the user touches the selection button **304**. This shifts the screen to the screen for balance inquiry. In accordance with the shifted screen, the user holds his/her card over the card reader **123**. Accordingly, user information including the balance held in the card is read by the card reader **123**, and the balance is displayed on the screen. Then, the user touches a button for returning the screen to the initial screen **300**. Accordingly, the initial screen **300** is displayed on the interface **121** again.

[0087] When the user wants to understand the method for using the locker apparatus **10** before depositing luggage, the user touches the selection button **305**. This shifts the screen to a screen for presenting the method for using the locker apparatus **10**. On this screen, the method for using the locker apparatus **10** is displayed as a moving image. Then, the user touches a button for returning the screen to the initial screen **300**. Accordingly, the initial screen **300** is displayed on the interface **121** again.

[0088] When the user has touched the selection button **302**, a message that urges the user to put luggage into a desired accommodation box **11** and temporarily lock the door **11a** is displayed on the interface **121**. In response to this, the user puts luggage into the accommodation box **11** selected by himself/herself among accommodation boxes **11** in an empty state, and temporarily locks the door **11a**. The locking device **212** transmits a detection signal of the temporary locking to the controller **221** of the settlement assembly **12** via the relay machine **240**. Accordingly, the controller **221** causes the interface **121** to display a confirmation screen **310** in FIG. 7.

[0089] In parallel with this, the controller **221** transmits a control signal for causing the lighting region **141** of the temporarily locked door **11a** to flicker, to the light emission device **211** for the lighting region **141** via the relay machine **240**. Accordingly, the lighting region **141** flickers.

[0090] The confirmation screen **310** includes a message **311**, a layout image **312**, and a cancel button **313**. The message **311** has a content that informs the user to lock the door **11a** of the accommodation box **11** displayed in green in the layout image **312**. Here, in the layout image **312**, the region (cell **312a**) of the hatched door **11a** having a locker number of "7" is shown in green. Further, the message **311** includes a content that urges the user to confirm the locker number and the position of the accommodation box **11** in which the luggage has been deposited and then touch the region (cell **312a**) of the door **11a** shown in green.

[0091] The layout image **312** is generated based on the above-described layout information and locker management information stored in the storage **222**. In the layout image **312**, the layout of a plurality of accommodation boxes **11** (doors **11a**) in the locker apparatus **10** are shown as cells **312a**. In addition, the layout image **312** includes an image **312b** showing the region of the settlement assembly **12** in the locker apparatus **10**. The cell **312a** of the accommodation box **11** corresponding to the locking device **212** having outputted a detection result of temporary locking is shown in green, and further, the locker number thereof is additionally shown.

[0092] When the green region (cell **312a**) displayed in the layout image **312** does not correspond to the accommodation box **11** in which the user has placed the luggage, the user touches the cancel button **313**. Accordingly, the controller **221** causes the interface **121** to display the initial screen **300** in FIG. 6. In this case, the user only needs to upwardly push the lever **143a** (see FIGS. 3A, 3B) of the accommodation box **11** in which the user has placed the luggage, to release the temporary locking of the door **11a** and take out the luggage from the accommodation box **11**. Then, the user may select another accommodation box **11** to which the luggage is to be accommodated, and perform the same step as above again.

[0093] When the user has been able to confirm that the green region (cell 312a) displayed in the layout image 312 corresponds to the accommodation box 11 in which the user has placed the luggage, the user touches the green region. Accordingly, the controller 221 causes the interface 121 to display a settlement screen 320 in FIG. 8.

[0094] The settlement screen 320 in FIG. 8 includes: a message 321 that urges selection of a payment method; a selection button 322 for selecting payment by an IC card; a selection button 323 for selecting payment by cash; a message 324 informing that, in the case of payment by the IC card, the IC card serves as the unlocking key; a message 325 informing that, in the case of payment by cash, a take-out ticket for taking out the luggage is to be issued; and a cancel button 326.

[0095] When the user performs settlement, by an IC card, of the payment for the day's use of the locker apparatus 10, the user touches the selection button 322. Accordingly, the controller 221 causes the interface 121 to display an IC card settlement screen. In accordance with this settlement screen, the user causes his/her IC card to be read by the card reader 123, to perform settlement.

When the settlement has been completed as a result of this, the controller 221 causes a use certificate to be issued through the first ticket issuing port 128a via the ticket issuing device 225.

[0096] At this time, the controller 221 stores the identification information of the IC card having been read, the identification information (locker number) of the accommodation box 11 serving as the settlement target this time, the day and time of the locking, and the key type indicating that the IC card is the key, in association with each other, into the storage 222. In addition, the controller 221 causes a corresponding locking device 212 to operate, to lock the door 11a of the accommodation box 11 serving as the settlement target this time.

[0097] When the user performs settlement of the payment, by cash, for the day's use of the locker apparatus 10, the user touches the selection button 323. Accordingly, the controller 221 causes the interface 121 to display the cash settlement screen. In accordance with this settlement screen, the user deposits banknotes or coins into the banknote inlet/outlet 124 or the coin inlet 125. When settlement has been completed as a result of this, the controller 221 causes a take-out ticket to be issued through the second ticket issuing port 128b via the ticket issuing device 225. This take-out ticket has printed thereon a two-dimensional barcode to serve as the key for unlocking the door 11a at the time of taking out the luggage.

[0098] The controller 221 stores the information of the issued two-dimensional barcode, the identification information (locker number) of the accommodation box 11 serving as the settlement target this time, the day and time of the locking, and the key type indicating that the two-dimensional barcode is the key, in association with each other, into the storage 222. In addition, the controller 221 causes a corresponding locking device 212 to operate, to lock the door 11a of the accommodation box 11 serving as the settlement target this time.

[0099] Then, the step of depositing luggage ends. The process performed when the cancel button 326 on the settlement screen 320 has been touched is the same as the process described above when the cancel button 313 in FIG. 7 has been touched.

<Taking Out of Luggage>

[0100] A flow of taking out luggage from the locker apparatus 10 will be described.

[0101] In this case, the user touches the selection button 303 on the initial screen 300 in FIG. 6.

Accordingly, the controller 221 causes the interface 121 to display a take-out screen 330 in FIG. 9.

[0102] The take-out screen 330 includes a message 331 that urges selection of a key type. As described above, when the user has performed settlement by an IC card on the settlement screen 320 in FIG. 8 (when the user has performed settlement by touching the selection button 322), the IC card serves as the key. When the user has performed settlement by cash on the settlement screen 320 (when the user has performed settlement by touching the selection button 323), the two-dimensional barcode printed on the take-out ticket serves as the key.

[0103] When the IC card is the key, the user touches a selection button 332 on the take-out screen 330. Accordingly, the controller 221 causes the interface 121 to display a screen including a

message that urges the user to cause the IC card to be read by the card reader **123**. In response to this, the user causes his/her IC card to be read by the card reader **123**.

[0104] The controller **221** checks the identification information of the read IC card against association information that is stored in the storage **222** and whose key type is IC card, and identifies the identification information (locker number) of the accommodation box **11** associated with the identification information. Then, the controller **221** causes the locking device **212** of the identified accommodation box **11** to execute unlocking of the door **11a**, and further performs control of causing the lighting region **141** in the panel **14** of the accommodation box **11** to flicker. When the luggage deposit period has been exceeded, the controller **221** performs settlement of the charge corresponding to the excess, from the IC card, and causes the interface **121** to display information indicating the same.

[0105] When the two-dimensional barcode is the key, the user touches a selection button **333** on the take-out screen **330**. Accordingly, the controller **221** causes the interface **121** to display a screen including a message that urges the user to cause the two-dimensional barcode on the take-out ticket to be read by the barcode reader **122**. In response to this, the user causes the two-dimensional barcode of his/her take-out ticket to be read by the barcode reader **122**.

[0106] The controller **221** checks the information of the read two-dimensional barcode against association information that is stored in the storage **222** and whose key type is two-dimensional barcode, and identifies the identification information (locker number) of the accommodation box **11** associated with the information of the two-dimensional barcode. Then, the controller **221** causes the locking device **212** of the identified accommodation box **11** to execute unlocking of the door **11a**, and further, performs control of causing the lighting region **141** in the panel **14** of the accommodation box **11** to flicker. When the luggage deposit period has been exceeded, the controller **221** causes, before execution of the unlocking, the interface **121** to display a screen for performing settlement of the charge corresponding to the excess and urges the user to perform settlement of the charge corresponding to the excess. In response to this, the user performs settlement by cash. In response to this, the controller **221** performs control of the unlocking and flicker of the lighting region **141** described above.

[0107] The user opens the unlocked door **11a** of the accommodation box **11** whose lighting region **141** is flickering, and takes out his/her luggage from the accommodation box **11**. After having taken out the luggage, the user closes the door **11a**. Accordingly, from the corresponding open/close sensor **213**, a detection result of opening/closing of the door **11a** is transmitted to the controller **221**. In response to this, the controller **221** performs control of setting the lighting region **141** of the accommodation box **11** to a lighting state. Then, the step of taking out the luggage ends.

[0108] In the take-out screen **330** in FIG. 9, a message **334** provided to the selection button **333** and a cancel button **335** are further included. Due to the message **334**, the user understands that, when having selected the selection button **333**, the user needs to perform settlement, by cash, for a payment corresponding to an exceeded period. When the user touches the cancel button **335**, the controller **221** causes the interface **121** to display the initial screen **300** in FIG. 6.

<Image Displaying Process>

[0109] Next, a process of displaying an advertisement image on the display **13** installed on the surface of a door **11a** will be described.

[0110] In the present embodiment, the controller **201** can execute two image display modes, i.e., a first mode and a second mode.

[0111] The first mode is a mode that is executed when the user has not performed an operation regarding opening of a door **11a**, and that is for providing an advertisement image to passersby walking in a passage in front of the locker apparatus **10**. The second mode is a mode that is executed when the user has performed an operation regarding opening of a door **11a**, and that is for providing an advertisement image to the user who uses the locker apparatus **10**.

[0112] Here, a fact that the user performs an operation of unlocking the door **11a** of the

accommodation box **11** he/she is using via the interface **121** at the time of taking out luggage, i.e., a fact that the user performs input for unlocking the door on the screens in FIG. **6** and FIG. **9**, is denoted as an “operation regarding opening of a door **11a**”.

[0113] When the controller **221** of the settlement assembly **12** has received an output according to an unlocking operation from the interface **121** and accordingly has unlocked the corresponding door **11a**, the controller **221** transmits a notification (unlocking completion notification) thereof, together with the identification information (the locker number of the accommodation box **11**) of the door **11a**, to the controller **201** of the control assembly **20**. Accordingly, the controller **201** executes the second mode described above.

[0114] When the controller **221** of the settlement assembly **12** has received a detection signal indicating that the door **11a** has been opened and then closed, from the open/close sensor **213** corresponding the unlocked door **11a**, the controller **221** transmits a notification (take-out completion notification) indicating the same, together with the identification information (the locker number of the accommodation box **11**) of the door **11a**, to the controller **201** of the control assembly **20**. Accordingly, the controller **201** ends the second mode and executes the first mode described above.

[0115] FIG. **10A** is a flowchart showing a display process of an advertisement image.

[0116] Until receiving the unlocking completion notification described above from the controller **221** of the settlement assembly **12** (**S102**: NO), the controller **201** executes a display process according to the first mode (**S101**). Upon receiving the unlocking completion notification, the controller **201** sets the determination in step **S102** to YES, and executes a display process according to the second mode (**S103**). Then, upon receiving the take-out completion notification described above (**S104**: YES), the controller **201** ends the process, and starts the process from step **S101** again. Accordingly, the display process according to the first mode is executed.

[0117] The take-out completion notification may be transmitted from the controller **221** of the settlement assembly **12** to the controller **201** in response to the fact that the door **11a** serving as the unlocking target has been opened. Alternatively, the take-out completion notification may be transmitted from the controller **221** of the settlement assembly **12** to the controller **201** in response to lapse of a predetermined time after the door **11a** serving as the unlocking target has been opened and then closed.

[0118] FIG. **10B** is a flowchart showing a process of the second mode in step **S103** in FIG. **10A**.

[0119] In accordance with the position (locker number) of the accommodation box **11** serving as the target of the unlocking operation, the controller **201** sets the doors **11a** to be used for displaying the advertisement image (**S201**), and causes the advertisement image to be displayed on the set doors **11a** (**S202**).

[0120] FIG. **11A** shows an example of the setting method of the doors **11a** to be used for displaying the advertisement image in the second mode. FIG. **11B** shows a setting example of a locker number with respect to each accommodation box **11**.

[0121] In the setting method in FIG. **11A**, in principle, a range of 3 vertically×3 horizontally doors **11a** including the door **11a** serving as the unlocking target at the center is set as the doors **11a** to be used for displaying the advertisement image. When this principle cannot be applied to the door **11a** serving as the unlocking target, the range (display range of the advertisement image) of 3 vertically×3 horizontally doors **11a** is set so as to include the door **11a** serving as the unlocking target at the center as much as possible.

[0122] For example, in FIG. **11B**, the door **11a** having a locker number of “1” is at the upper left corner, and thus, if this door **11a** is set at the center of the display range, the advertisement image to be displayed in the range of 3 vertically×3 horizontally doors **11a** can be displayed only in the range of the doors **11a** having locker numbers of “1, 2, 5, 6”. Therefore, in the setting example in FIG. **11A**, when the door **11a** having a locker number of “1” is the unlocking target, a range (the range of locker numbers of “1 to 3, 5 to 7, 9 to 11”) of 3 vertically×3 horizontally doors **11a**

including this door **11a** at the upper left corner is set as the range of the doors **11a** to be used for displaying the advertisement image.

[0123] Similarly, when the door **11a** having a locker number of “2” is the unlocking target, a range (the range of locker numbers of “1 to 3, 5 to 7, 9 to 11”) of 3 vertically×3 horizontally doors **11** including this door **11a** at the center on the left side is set as the range of the doors **11a** to be used for displaying the advertisement image.

[0124] Here, since the display range of the advertisement image in the second mode is set to the range of 3 vertically×3 horizontally doors **11a**, when the door **11a** having a locker number of “21 to 26” is the unlocking target, the range of 3 vertically×3 horizontally doors **11a** cannot be set so as to include the door **11a**. Therefore, as shown in FIG. **11A**, when any one of the doors **11a** having locker numbers of “21, 23, 24” is the unlocking target, the range of the doors **11a** having locker numbers of “9 to 11, 13 to 15, 17 to 19” in the vicinity of these doors **11a** is set as the range of the doors **11a** to be used for displaying the advertisement image. When any one of the doors **11a** having locker numbers of “22, 25, 26” is the unlocking target, the range of the doors **11a** having locker numbers of “10 to 12, 14 to 16, 18 to 20” in the vicinity of these doors **11a** is set as the range of the doors **11a** to be used for displaying the advertisement image.

[0125] In this setting method, the displays **13** of the doors **11a** having locker numbers of “21 to 26” are not used for displaying the advertisement image. However, the range of the doors **11a** to be used for displaying the advertisement image is not limited to the range of 3 vertically×3 horizontally doors **11a**. For example, when the advertisement image is displayed by using only the range of one door **11a**, the advertisement image may be displayed by using only the door **11a** serving as the unlocking target. In this case, the displays **13** of the doors **11a** having locker numbers of “21 to 26” can also be used for displaying the advertisement image.

[0126] FIGS. **12A** to **12C** each schematically show display methods of the advertisement image in the first mode and the second mode.

[0127] In FIGS. **12A** to **12C**, the upper row shows a display method of the advertisement image according to the first mode, and the lower row shows a display method of the advertisement image according to the second mode. Here, the range of 16 doors **11a** of 4 vertically×4 horizontally is set as a display range **A1** for the advertisement image in the first mode, and the range of nine doors **11a** of 3 vertically×3 horizontally is set as a display range **A2** for the advertisement image in the second mode. For convenience, each advertisement image is shown as an image of a character **P** to **S**. In the following, description will be given assuming that the advertisement images are advertisement images **P**, **Q**, **R**, **S**, respectively.

[0128] In the example in FIG. **12A**, one advertisement image **P** is displayed in each mode. Since the display range **A2** in the second mode is smaller than the display range **A1** in the first mode, the advertisement image **P** in the second mode is reduced as compared with the first mode.

[0129] In the example in FIG. **12B**, four advertisement images **P**, **Q**, **R**, **S** are displayed in each mode. Here as well, due to the size relationship between the display ranges **A1**, **A2**, the advertisement image **P** in the second mode is reduced as compared with the first mode.

[0130] In the example in FIG. **12C**, four advertisement images **P**, **Q**, **R**, **S** are displayed in the first mode, and only the advertisement image **P** is displayed in the second mode. Here, since the number of advertisement images that are displayed is smaller in the second mode than in the first mode, the advertisement image **P** in the second mode is enlarged as compared with the first mode. In this case, the advertisement image displayed in the second mode may be cyclically transitioned in the order of **P**, **Q**, **R**, **S** every predetermined time.

[0131] The display method of the advertisement image in the first mode and the second mode is not limited to the examples shown in FIGS. **12A** to **12C**. For example, in the examples in FIGS. **12A** to **12C**, at least parts of the displayed advertisement images overlap between the first mode and the second mode. However, completely different advertisement images may be displayed between the first mode and the second mode. The range (number) of the doors **11a** to be used for displaying in

the first mode and the second mode is not limited to the examples shown in FIGS. 12A to 12C, either. In accordance with change in the range (number) of the doors 11a to be used for displaying, compression or enlargement of the advertisement image may be adjusted. In either mode, the advertisement image may be displayed at a size that allows the advertisement image to be easily seen by passersby or the user, while the display ranges A1, A2 are efficiently used.

[0132] The manager may be allowed to set, via the interface 121, which of the methods in FIGS. 12A to 12C is used to display the advertisement image. Alternatively, the manager of the locker apparatus 10 may be allowed to arbitrarily set, via the interface 121, a display method (the setting method of the display ranges A1, A2, the method of assigning the advertisement image to each display range, etc.) of the advertisement image in the first mode and the second mode.

[0133] As described above, display of the advertisement image is performed by using the displays 13 of the doors 11a included in the display ranges A1, A2. Therefore, the controller 201 controls each display 13 included in the display ranges A1, A2 such that the advertisement image is displayed in the form of FIGS. 12A to 12C over the entirety of the display ranges A1, A2. At this time, the controller 201 assigns, to each display 13, image information of the advertisement image excluding the portion of the advertisement image corresponding to the gap between adjacent displays 13, and drives each display 13. Therefore, the advertisement image that is displayed is an image lacking in the portions between the gaps. However, since the proportion of the area of this portion relative to the entire area is small, passersby or the user can smoothly refer to the advertisement image.

[0134] FIG. 13 shows a setting example of the display range A1 in the first mode.

[0135] In the example in FIG. 13, the range of the doors 11a in the first column to the fourth column from the left is set as the display range A1 for the advertisement image in the first mode. The range of the doors 11a in the second column to the fifth column from the left may be set as the display range A1 for the advertisement image in the first mode. Alternatively, when the display range A1 in the first mode is the range of the doors 11a of 4 vertically×5 horizontally, the range of the doors 11a in the first column to the fifth column from the left may be set as the display range A1 for the advertisement image in the first mode.

[0136] FIG. 14 to FIG. 17 each show a setting example of the display range A2 in the second mode.

[0137] The setting examples in FIG. 14 to FIG. 17 are according to the setting method in FIG. 11A. In FIG. 14 to FIG. 17, the hatched door 11a is the door 11a for which an unlocking operation has been performed. For convenience, the hatched door 11a is provided with a locker number.

[0138] As shown in FIG. 14, when the range of the doors 11a of the 3 vertically×3 horizontally having the door 11a serving as the unlocking target at the center can be set, this range is set as the display range A2 for the advertisement image in the second mode. As shown in FIG. 15 to FIG. 17, when the range of the doors 11a of 3 vertically×3 horizontally having the door 11a serving as the unlocking target at the center cannot be set, the range of the doors 11a of 3 vertically×3 horizontally including the door 11a serving as the unlocking target is set as the display range A2 for the advertisement image in the second mode. In this case, as shown in FIG. 15 and FIG. 16, the display range A2 for the advertisement image is set such that the door 11a serving as the unlocking target is as close as possible to the center of the display range A2. When the door 11a serving as the unlocking target cannot be made close to the center of the display range A2, the display range A2 for the advertisement image is set so as to include the door 11a serving as the unlocking target at a corner of the display range A2, as shown in FIG. 17.

[0139] Such a setting of the display range A2 can be realized by a method in which an algorithm according to the setting method (setting rule) in FIG. 11A is stored in the storage 202. In this case, the controller 201 identifies the position of the accommodation box 11 (door 11a) serving as the unlocking target, on the layout of the accommodation boxes 11 based on the layout information, and applies the identified position and the layout to the algorithm, to set the display range A2 for

the advertisement image.

[0140] Alternatively, setting of the display range **A2** can also be realized by a method in which a table similar to that in FIG. **11A** is stored in the storage **202**. In this case, the controller **201** sets the range of the doors **11a** having locker numbers associated with the locker number of the door **11a** serving as the unlocking target in the table, as the display range **A2** for the advertisement image. The table may be set in advance by the manager or the installer of the locker apparatus **10** in accordance with the layout of the accommodation boxes **11**. This setting may be performed via the interface **121**, or may be performed by connecting a dedicated terminal to the control assembly **20**.
Effects of Embodiment 1

[0141] As shown in FIG. **1**, FIG. **2**, and FIG. **5**, the accommodation system **1** includes: a plurality of the accommodation boxes **11** each having a door **11a** and disposed so as to be arranged in an in-plane direction; the display **13** (display device) that displays an image on each door **11a**; and the controller **201** that acquires, via the interface **121**, information (a locker number of “7” in the example in FIG. **7**) on the accommodation box **11** for which an operation (unlocking operation) regarding opening of a door **11a** has been performed, sets the doors **11a** (the display range **A2**) on which an advertisement image is to be displayed, in accordance with the position of the accommodation box **11** identified by the outputted information, and causes the display **13** (display device) to display the advertisement image on the set doors **11a**.

[0142] With this configuration, in accordance with the position of the accommodation box **11** for which the operation (unlocking operation) regarding opening of a door **11a** has been performed by the user, the doors **11a** for displaying the advertisement image are set. Therefore, the user can naturally refer to the advertisement image at the time of his/her use of the accommodation box **11**.

[0143] As shown in FIG. **1**, FIG. **2**, and FIG. **5**, each door **11a** is configured to be able to be locked by the locking device **212**, the locker apparatus **10** includes the interface **121** that receives an operation of unlocking the door **11a** for each accommodation box **11**, and the interface **121** outputs information (the locker number of “7” in the example in FIG. **7**) that identifies the accommodation box **11** for which the operation of unlocking has been performed on the interface **121**, as the information on the accommodation box **11** for which the operation regarding opening of a door **11a** has been performed.

[0144] With this configuration, in accordance with the position of the accommodation box **11** for which the operation of unlocking has been performed by the user via the interface **121**, the doors **11a** for displaying the advertisement image are set. Therefore, the user can naturally refer to the advertisement image when advancing toward the accommodation box **11** in which his/her luggage is accommodated, after performing an operation of unlocking on the interface **121**.

[0145] As shown in FIG. **14** to FIG. **17**, in the second mode, the controller **201** causes the door **11a** of the accommodation box **11** for which the operation (unlocking operation) regarding opening of a door **11a** has been performed, to be included in the doors **11a** on which the advertisement image is to be displayed.

[0146] With this configuration, since the advertisement image is displayed on the very door **11a** of the accommodation box **11** used by the user, the user can smoothly refer to the advertisement image in the second mode.

[0147] As shown in FIG. **14** to FIG. **17**, in the second mode, the controller **201** causes predetermined doors **11a** in the vicinity of the accommodation box **11** for which the operation (unlocking operation) regarding opening of a door **11a** has been performed, to be included in the doors **11a** on which the advertisement image is to be displayed.

[0148] With this configuration, since an image is displayed on the doors **11a** in the vicinity of the accommodation box **11** used by the user, the user can smoothly refer to the advertisement image in the second mode.

[0149] As shown in FIG. **14** to FIG. **17**, in the second mode, the controller **201** causes the doors **11a** of predetermined accommodation boxes **11** adjacent to the accommodation box **11** for which

the operation (unlocking operation) regarding opening of a door **11a** has been performed, to be included in the doors **11a** on which the advertisement image is to be displayed.

[0150] With this configuration, since an image is displayed on the doors adjacent to the accommodation box used by the user, the user can smoothly refer to the advertisement image in the second mode.

[0151] As shown in FIG. **10A**, when the operation (unlocking operation) regarding opening of a door **11a** has not been performed (**S102**: NO), the controller **201** executes the first mode (**S101**) to cause the advertisement image to be displayed on predetermined doors **11a**. At this time, as shown in FIGS. **12A** to **12C**, the controller **201** causes the advertisement image displayed when the operation has not been performed (the first mode) and the advertisement image displayed when the operation has been performed (the second mode) to be different from each other.

[0152] With this configuration, when there is no user who uses an accommodation box **11**, an advertisement image for passersby is displayed according to the first mode, and when there is a user who uses an accommodation box **11**, an advertisement image for the user is displayed according to the second mode. Therefore, it is possible to allow not only the user but also passersby to smoothly refer to the advertisement image.

[0153] As shown in FIGS. **12A** to **12C**, the controller **201** decreases, as compared with the number of the doors **11a** on which the advertisement image is to be displayed when the operation (unlocking operation) regarding opening of a door **11a** has not been performed (the first mode), the number of the doors **11a** on which the advertisement image is to be displayed when the operation has been performed (the second mode).

[0154] With this configuration, when there is no user who uses an accommodation box **11**, an advertisement image having a large size that allows the advertisement image to be easily seen by passersby at a position away from the doors **11a** can be displayed, and when there is a user who uses an accommodation box **11**, an advertisement image having a small size that allows the advertisement image to be easily seen by the user at a position close to the doors **11a** can be displayed. Accordingly, in accordance with the respective situations, it is possible to allow the user and passersby to smoothly refer to the advertisement images, respectively.

[0155] As shown in FIG. **2** and FIG. **5**, the advertisement image is displayed by the display **13** disposed on the surface of the door **11a**.

[0156] With this configuration, by adjusting the display **13** that displays the advertisement image, in accordance with the position of the accommodation box **11** used by the user, the advertisement image can be displayed on the doors **11a** serving as the display target.

[0157] As described above, the image displayed on the door **11a** is an advertisement image including advertisement information.

[0158] With this configuration, the advertisement image can be smoothly displayed to the user who uses the accommodation box **11**, and an advertisement effect on the user can be enhanced.

Modification 1

[0159] In Embodiment 1 above, in principle, the display range **A2** for the advertisement image in the second mode is set so as to have the door **11a** serving as the unlocking target at the center. However, the setting method of the display range **A2** is not limited thereto.

[0160] For example, as shown in FIG. **18**, the display range **A2** may be set in the vicinity of the door **11a** serving as the unlocking target, such that the door **11a** adjacent in the opening direction (here, the left direction since the door **11a** is left-opening) to the door **11a** serving as the unlocking target is not included in the display target.

[0161] When the display range **A2** is set like this, the left-adjacent door **11a** hidden by the opened door **11a** is excluded from the display target of the advertisement image. Therefore, the user can smoothly refer to the advertisement image.

[0162] As shown in FIG. **19**, the display range **A2** may be set in the vicinity of the door **11a** serving as the unlocking target such that the door **11a** serving as the unlocking target is not included in the

display target. Here, the display range A2 is set such that the door **11a** adjacent, in the direction opposite to the opening direction, to the door **11a** serving as the unlocking target is included in the display target.

[0163] When the door **11a** serving as the unlocking target is opened, the range of the advertisement image corresponding to this door **11a** becomes difficult to be seen. In contrast, when the display range A2 is set as in FIG. **19**, even when the door **11a** serving as the unlocking target is opened, the advertisement image being displayed in the display range A2 is not affected. Therefore, the user can smoothly refer to the advertisement image.

[0164] As shown in FIG. **20**, as long as the display range A2 in the second mode is in the vicinity of the door **11a** serving as the unlocking target, the display range A2 in the second mode need not necessarily include the door **11a** adjacent to this door **11a**.

[0165] When the door **11a** serving as the unlocking target is in the lowest row of the locker apparatus **10** as in FIG. **20**, the user performs an unlocking operation on the settlement assembly **12**, then moves to the column of the door **11a** serving as the unlocking target, and then, crouches and opens the door **11a** serving as the unlocking target. In this case, it can be assumed that, while the user moves from the settlement assembly **12** to the column of the door **11a** serving as the unlocking target, it is easier to see the advertisement image, due to the relationship with a general height, if the display range A2 (here, the display range A2 is set to a range of the doors **11a** of 2 vertically×2 horizontally) is set in an upper part as in FIG. **20**. Therefore, if the display range A2 is set like this, the user can smoothly refer to the advertisement image.

Modification 2

[0166] In Embodiment 1 above, the display range A2 in the second mode is fixed. However, in Modification 2, the display range A2 is re-set in accordance with opening of the door **11a** serving as the unlocking target, and the advertisement image is displayed in the re-set display range A2.

[0167] FIG. **21** is a flowchart showing a process of the second mode according to Modification 2.

[0168] In the flowchart in FIG. **21**, steps S211 to S213 are added as compared with the flowchart in FIG. **10B**. Until the door **11a** serving as the unlocking target is opened (S211: NO), the controller **201** causes the advertisement image to be displayed while, in principle, the display range A2 is set to the range of the doors of 3 vertically×3 horizontally having the door **11a** serving as the unlocking target at the center (S202), as in Embodiment 1 above.

[0169] Then, when the user has moved to the position of the door **11a** serving as the unlocking target and has opened the door **11a**, an output indicating the same from the open/close sensor **213** is transmitted to the controller **221** of the settlement assembly **12**. The controller **221** transmits a notification corresponding to this output, together with the identification information (locker number) of the opened door **11a**, to the controller **201** of the control assembly **20**. Due to this notification, the controller **201** recognizes that the door **11a** serving as the unlocking target has been opened (S211: YES).

[0170] Accordingly, the controller **201** re-sets the range (the display range A2) of the doors **11a** to serve as the display target (S212), and causes the advertisement image to be displayed in the re-set range of the doors **11a** (S213). In step S212, the controller **201** re-sets, so as to exclude the opened door **11a** from the display target, a new display range A2 in the vicinity of the door **11a**.

[0171] FIG. **22** shows the re-set range (the display range A2) of the doors **11a**.

[0172] Here, with respect to the opened door **11a** (here, the door **11a** having a locker number of “7”), a range of the doors **11a** of 2 vertically×2 horizontally including the door **11a** adjacent in the direction (right direction) opposite to the opening direction (left direction) of the door is re-set as the display range A2. In this case, the controller **201** causes the advertisement image reduced in accordance with the size of the re-set display range A2, to be displayed on four displays **13** included in the display range A2. The display range A2 after the re-setting may have a size of 3 vertically×3 horizontally as before the re-setting.

[0173] As described above, in Modification 2, after an operation of unlocking thereof has been

performed on the interface **121** and until opening of the door **11a** is performed, the controller **201** causes the door **11a** serving as the unlocking target to be included in the display target of the advertisement image, and after the door **11a** has been opened, the controller **201** causes the door **11a** serving as the unlocking target to be excluded from the display target of the advertisement image.

[0174] With this configuration, when the user advances toward the accommodation box **11** in which his/her luggage is accommodated, after performing the operation of unlocking on the interface **121**, the advertisement image is displayed on the doors **11a** including the accommodation box **11** being used by the user himself/herself, and thus, the user can smoothly refer to the advertisement image. Thereafter, when the user has opened the door **11a** of the accommodation box **11**, this door **11a** is excluded from the display target of the advertisement image. Therefore, it is possible to avoid the advertisement image being continuously displayed on the door **11a** that has been opened and has become difficult to appropriately provide the advertisement image. In this case as well, in the vicinity of the position of the accommodation box **11** used by the user, the doors **11a** for displaying the advertisement image are re-set, and thus, the user can smoothly refer to the advertisement image also after the door **11a** has been opened.

Modification 3

[0175] In Embodiment 1 above, the “operation regarding opening of a door”, which is an execution condition for the second mode, is “that an operation of unlocking a door **11a** has been performed on the interface **121**”. However, the “operation regarding opening of a door”, which is the execution condition for the second mode, is not limited thereto and may be another operation associated with opening of a door **11a**.

[0176] For example, an operation, performed by the user, of gripping the handle **11b** and opening the door **11a** at the time of taking out or depositing luggage may be the “operation regarding opening of a door”. Alternatively, an operation, performed by the user, of opening a door **11a** and then closing the door **11a** at the time of depositing luggage may be the “operation regarding opening of a door”, or further, after this operation, an operation of pushing and lowering the lever **143a** of the temporary lock operation unit **143** to temporarily lock the door **11a** may be the “operation regarding opening of a door”. In this case, a notification (temporary locking notification) that the temporary locking has been performed is transmitted together with the locker number of the temporarily locked door **11a**, from the controller **221** to the controller **201**. Thus, the “operation regarding opening of a door”, which is the execution condition for the second mode, may be an operation that can reflect that the user moves to the position of the door **11a** serving as an opening target or is at this position.

[0177] FIG. **23A** is a flowchart showing a display process of an advertisement image according to Modification 3. FIG. **23B** is a flowchart showing a process of the second mode in step **S103** in FIG. **23A**.

[0178] In the flowchart in FIG. **23A**, steps **S102**, **S104** in the flowchart in FIG. **10A** are replaced with steps **S111**, **S112**. In the flowchart in FIG. **23B**, step **S201** in the flowchart in FIG. **10B** is replaced with step **S221**.

[0179] Here, a fact that a certain door **11a** has been detected to be opened by the open/close sensor **213** is the execution condition for the second mode. In this case, in response to reception of a detection signal indicating opening of the door **11a** from the open/close sensor **213**, the controller **221** of the settlement assembly **12** transmits a notification (opening notification) indicating the same, together with the identification information of the opened door **11a**, to the controller **201** of the control assembly **20**. In response to this, the controller **201** executes the second mode. Then, in response to reception from the open/close sensor **213** of a detection signal indicating that the door **11a** has been closed, the controller **221** of the settlement assembly **12** transmits a notification (closing notification) indicating the same, together with the identification information of the closed door **11a**, to the controller **201** of the control assembly **20**. In response to this, the controller **201**

ends the second mode and executes the first mode.

[0180] With reference to FIG. 23A, upon reception of the opening notification described above (S111: YES), the controller 201 executes the second mode (S103). Then, upon reception of the closing notification (S112: YES), the controller 201 ends the process of the flowchart, and starts the process from step S101 again. Accordingly, the display process in the first mode is executed.

[0181] With reference to FIG. 23B, the controller 201 sets the doors 11a (the display range A2) to serve as the display target, in accordance with the position of the opened accommodation box 11 (S221). This setting is performed similarly to Embodiment 1 above. That is, here, the door 11a serving as the reference is merely changed from the door 11a serving as the unlocking target to the door 11a serving as an opening target. As a setting method of the doors 11a to serve as the display target, the setting method in Modification 1 above may be used. The controller 201 causes the display 13 on the set doors 11a (the display range A2) to display the advertisement image (S202).

[0182] As described above, the accommodation system 1 according to Modification 3 includes the open/close sensor 213 that detects opening of a door 11a, and the open/close sensor 213 outputs a detection result corresponding to the opening of the door 11a, as information on the accommodation box 11 for which the operation regarding opening of a door 11a has been performed.

[0183] With this configuration, in accordance with the position of the accommodation box 11 opened by the user, the doors 11a for displaying the advertisement image are set. Therefore, after opening the door 11a, the user can naturally refer to the advertisement image.

[0184] In FIG. 23A, as another example of the “operation regarding opening of a door”, a fact that a door 11a has been opened (S111) has been shown as an example. However, in said another case of the “operation regarding opening of a door” described above as well, steps S111, S112 may be modified as appropriate.

[0185] For example, when the “operation regarding opening of a door” is that a door 11a has been opened and then closed, step S111 may be changed to a determination condition (presence or absence of reception of the closing notification) corresponding to this, and for step S112, a fact that, after this determination condition has been satisfied, a predetermined time (a time required for the user to refer to the advertisement image: e.g. several seconds to about 10 seconds) has elapsed, may be set as the determination condition.

[0186] When the “operation regarding opening of a door” is that a door 11a has been temporarily locked, step S111 may be changed to a determination condition (presence or absence of reception of the temporary locking notification) corresponding to this, and for step S112, a fact that, after this condition has been satisfied, a predetermined time (a time required for the user to refer to the advertisement image: e.g. several seconds to about 10 seconds) has elapsed, may be set as the determination condition.

Modification 4

[0187] In Embodiment 1 above, when the “operation regarding opening of a door” (in Embodiment 1, an unlocking operation performed on the interface 121), which is the execution condition for the second mode, has been performed, the second mode is always executed. In contrast, in Modification 4, whether or not to execute the second mode can be set by the manager. That is, the controller 201 receives whether or not to execute the second mode from the manager of the accommodation system 1 via the interface 121. Then, the controller 201 stores, into the storage 202, necessity/unnecessity information indicating necessity/unnecessity of execution of the second mode received from the manager.

[0188] The necessity/unnecessity information may be initially set to a state where the second mode is executed. In this case, when execution of the second mode is not necessary, the manager may cause the interface 121 to display a setting screen, to change the necessity/unnecessity information to a state where the second mode is not executed.

[0189] FIG. 24 is a flowchart showing a display process of an advertisement image according to

Modification 4.

[0190] In the flowchart in FIG. 24, step S121 is added as compared with the flowchart in FIG. 10A. When the controller 201 has received an unlocking operation via the interface 121 (S102: YES), the controller 201 refers to the necessity/unnecessity information and determines whether or not a setting of executing the second mode has been made (S121). When the setting of executing the second mode has been made (S121: YES), the controller 201 advances the process to step S103, and executes the same process as in FIG. 10A. When the setting of executing the second mode has not been made (S121: NO), the controller 201 ends the process of this time, and starts the process from step S101 again.

[0191] With this configuration, the manager of the accommodation system 1 can arbitrarily set whether or not to execute the second mode in which the advertisement image for the user is displayed. Therefore, convenience for the manager can be enhanced.

[0192] In Modification 4 as well, Modification 2 may be applied, and the determination conditions in steps S102, S104 may be changed as in Modification 3.

Embodiment 2

[0193] In Embodiment 1 above, the advertisement image is displayed by the display 13 installed on the surface of the door 11a, but the configuration for causing the advertisement image to be displayed on the door 11a is not limited thereto. For example, the advertisement image may be displayed on the door 11a by projecting the advertisement image onto the door 11a.

[0194] FIG. 25 shows a configuration of the accommodation system 1 according to Embodiment 2.

[0195] In Embodiment 2, the display 13 is omitted from each door 11a, and a projector 50 is added. The projector 50 is disposed so as to be able to project an image onto the surface of the door 11a. The projection range of the projector 50 generally matches the entire surface of the locker apparatus 10. Therefore, the projector 50 can project an image onto the surfaces of all the doors 11a disposed in the locker apparatus 10. The projector 50 is communicably connected to the control assembly 20.

[0196] The projector 50 projects an advertisement image through control from the control assembly 20 (the controller 201). The controller 201 controls the projector 50 such that the advertisement image is displayed on a predetermined door 11a in the same form as in Embodiment 1 above.

[0197] That is, in the first mode, the controller 201 controls the projector 50 such that the advertisement image is projected to be displayed in the display range A1 including the doors 11a of 4 vertically×4 horizontally, in the projection range. In the second mode, the controller 201 controls the projector 50 such that the advertisement image is projected to be displayed in the display range A2 including the doors 11a of 3 vertically×3 horizontally in the vicinity of the door 11a serving as the unlocking target, in the projection range. Switching between the first mode and the second mode is performed in accordance with the process in FIG. 10A as in Embodiment 1 above. In Embodiment 2 as well, Modifications 1 to 4 above may be similarly applied.

Effects of Embodiment 2

[0198] In Embodiment 2 as well, effects similar to those in Embodiment 1 and modifications thereof are exhibited.

[0199] With the configuration of Embodiment 2, when the projection range of the advertisement image by the projector 50 is adjusted in accordance with the position of the accommodation box 11 used by the user, the advertisement image can be displayed on the doors 11a serving as the display target.

Other Configuration Examples

[0200] In Embodiments 1, 2 above, control of the second mode is performed such that the advertisement image is displayed on the doors 11a in the vicinity of the accommodation box 11 for which the user has performed an operation regarding opening of a door 11a. In contrast, in another configuration example, control of the second mode is performed such that the position of the user who is going to use the locker apparatus 10 is detected and the advertisement image is displayed on

the doors **11a** in the vicinity of this position.

[0201] FIG. **26** shows a configuration of the accommodation system **1** according to another configuration example.

[0202] Here, a camera **60** is added to the configuration in FIG. **1**. The camera **60** is communicably connected to the control assembly **20**. The camera **60** captures an image of a region in front of the locker apparatus **10**. Here, the camera **60** is installed at the top face of the locker apparatus **10**. The camera **60** may be installed at the ceiling above the locker apparatus **10**.

[0203] The range of the field of view of the camera **60** is set so as to be able to take an image of the user present at a position away, by a predetermined distance (e.g., about 0.5 to 1 m) from each door **11a** of the locker apparatus **10**. When this setting is difficult to be made with a single camera **60**, a plurality of the cameras **60** may be installed. For example, one camera **60** may be installed for the doors **11a** corresponding to n columns from the left, and another camera **60** may be installed for the doors **11a** corresponding to the $(n+1)$ th column to the column at the right end.

[0204] The storage **202** of the control assembly **20** stores therein an analytical engine for analyzing a taken image from the camera **60**. The controller **201** of the control assembly **20** analyzes the taken image from the camera **60** by this analytical engine, and determines whether or not the user faces any one of the doors **11a**. More specifically, the controller **201** extracts the face of the user from the taken image, and determines whether or not the extracted face is stationary in a state of facing the locker apparatus **10**. When the face has become stationary in a state of facing the locker apparatus **10**, the controller **201** identifies the accommodation box **11** that the user faces, from the position of the face on the taken image.

[0205] The controller **201** executes the second mode described above in response to the fact that the face of the user has become stationary while facing the locker apparatus **10**. That is, in this configuration example, the determination condition in step **S102** in FIG. **10A** is changed to whether or not the face of the user is stationary while facing the locker apparatus **10**. In the second mode, the controller **201** sets the range of the doors **11a** near the accommodation box **11** identified from the position of the face, as the display range **A2**. That is, in this configuration example, the “accommodation box **11** serving as the target of the unlocking operation” in step **S201** in FIG. **10B** is changed to the “accommodation box identified from the position of the face” described above. The controller **201** causes the displays **13** of the doors **11a** in the set display range **A2** to display the advertisement image. This display is performed by the same process as that in Embodiment 1 above.

[0206] Then, the controller **201** continues the analysis of the taken image, and determines whether or not the user has moved from the “accommodation box **11** identified” described above. When having determined that this movement has occurred, the controller **201** ends the second mode and executes the first mode. That is, the determination condition in step **S104** in FIG. **10A** is changed to whether or not the user has moved from the “accommodation box **11** identified” described above.

[0207] With this configuration example as well, effects similar to those in Embodiment 1 above are exhibited. With this configuration example as well, change similar to that in Modifications 1, 2, 4 above may be performed. Further, in this configuration example as well, as in Embodiment 2 above, the advertisement image may be displayed by the projector **50**.

[0208] The method of determining whether or not the user is present near the front of the locker apparatus **10** is not limited to the above method. For example, based on a fact that the face or the head has simply become substantially stationary on the taken image even when the user does not face the locker apparatus **10**, the second mode may be executed. In this case as well, from the position of the face or the head on the taken image, the accommodation box **11** corresponding to this position may be identified.

[0209] Detection of the position of the user may be performed by using an apparatus other than the camera **60**. For example, for each column of the doors **11a** in the locker apparatus **10**, a human sensor having a narrow directivity according to the width of the column may be disposed, and

based on a fact that the human sensor has continuously detected a person for a certain period, it may be determined that there is a user at the corresponding column. In this case, for example, based on the second accommodation box **11** (the position of the face of the user in a standing state) from the top in this column, the display range **A2** in the second mode may be set. Further, in accordance with a fact this human sensor has not detected a person anymore, the second mode may be ended, and the mode may be shifted to the first mode.

Other Modifications

[0210] In Embodiment 1 above, in step **S104** in FIG. **10A**, based on a fact that an operation of taking out luggage from a corresponding accommodation box **11** has been performed, the second mode is ended. However, the ending condition for the second mode is not limited thereto. For example, based on a fact that a predetermined time (e.g., several tens of seconds to about one minute) has elapsed from unlocking of a door **11a**, the second mode may be ended. Alternatively, when the camera **60** is separately installed as in FIG. **26**, the second mode may be ended when it has been determined that the user has gone away from the door **11a** serving as the unlocking target, based on a taken image from the camera **60**. This also applies to Embodiment 2 above.

[0211] In Embodiments 1, 2 above, in the second mode, the advertisement image is displayed only in the display range **A2**. However, during execution of the second mode, also in a range other than the display range **A2** in the second mode, another advertisement image may be further displayed, for example, for passersby.

[0212] The size of the display range **A2** in the second mode is not limited to the sizes shown in Embodiments 1, 2 and the modifications described above. For example, in the second mode, the advertisement image may be displayed only on the door **11a** serving as the unlocking target, or the advertisement image may be displayed only on one door **11a** in the vicinity of the door **11a** serving as the unlocking target. The size of the display range **A2** in the second mode may be set to a size that allows the display range **A2** to be easily seen by the user who uses the accommodation box **11**.

[0213] When an IC card includes attribute information (age, sex, etc.) of the user, the advertisement image to be displayed in the second mode may be changed in accordance with the attribute information read from the IC card. This attribute information may be acquired through analysis of a taken image from a camera. For example, when the settlement assembly **12** has a camera, from a taken image of the user facing the settlement assembly **12** at the time of depositing or taking out luggage, attribute information of the user may be acquired.

[0214] In Embodiments 1, 2 above, the control assembly **20** is a personal computer built in the locker apparatus **10**. However, the control assembly **20** may be a personal computer installed separately from the locker apparatus **10**, and may perform the control described above via the external communication network **40**. The control assembly **20** used in common between a plurality of the locker apparatuses **10** may perform the control described above.

[0215] Alternatively, in the configuration in Embodiment 1 above, the function of the controller **201** of the control assembly **20** may be implemented in the controller **221** of the settlement assembly **12**, and the control assembly **20** may be omitted. In this case, the accommodation system **1** is formed by the single locker apparatus **10**. In the configuration in Embodiment 2 above as well, the function of the controller **201** of the control assembly **20** may be implemented in the controller **221** of the settlement assembly **12**, and the control assembly **20** may be omitted. In this case, the projector **50** is communicably connected to the settlement assembly **12**.

[0216] The configuration of the locker apparatus **10** shown in Embodiments 1, 2 above, and the layout of the accommodation boxes **11** and the doors **11a** are examples, and another configuration or layout may be adopted. The size of the accommodation box **11** (door **11a**) is also an example, and an accommodation box **11** (door **11a**) having a size different from those in Embodiments 1, 2 may be adopted. The sizes of the accommodation boxes **11** (door **11a**) need not be uniform either. Several types of accommodation boxes **11** (doors **11a**) having different sizes may be disposed in the locker apparatus **10**. In accordance with the layout and size of the doors **11a**, the display range

A1 in the first mode and the display range A2 in the second mode can be changed as appropriate.

The doors 11a may be right-opening.

[0217] In Embodiments 1, 2 above, the first mode and the second mode can be executed. However, the first mode may be omitted and only the second mode may be able to be executed.

[0218] In Embodiments 1, 2 above, the image displayed in the second mode is an advertisement image. However, the image displayed in the second mode may be an image other than the advertisement image. For example, when the locker apparatus 10 is installed in a station, an announcement image regarding stations and trains such as a train operation status may be displayed in the second mode. Alternatively, an entertainment image, such as an image of a dog, a cat, and the like, that entertains the user of the locker apparatus 10 may be displayed in the second mode. Accordingly, an announcement image or an entertainment image can be smoothly displayed to the user, and information useful for the user or an image that entertains the user can be provided to the user.

[0219] In addition to the above, modification can be made as appropriate to the embodiments of the present invention within the scope defined by the claims.

Claims

1. An accommodation system comprising: a plurality of accommodation boxes each having a door and disposed so as to be arranged in an in-plane direction; a display device configured to display an image on a door of an accommodation box of the plurality of accommodation boxes; and a controller configured to acquire information on the accommodation box for which an operation of opening the door has been performed, set the door, on which the image is to be displayed, in accordance with a position of the accommodation box identified by the acquired information, and control the display device to display the image on the door.
2. The accommodation system according to claim 1, wherein each door is configured to be able to be locked, the accommodation system further comprises an interface configured to receive an operation of unlocking a door of an accommodation box of the plurality of accommodation boxes, and the interface outputs identification information that identifies the accommodation box for which the operation of unlocking has been performed on the interface, as the information on the accommodation box for which the operation of opening the door has been performed.
3. The accommodation system according to claim 2, wherein after the operation of unlocking has been performed on the interface and until opening the door has been performed, the controller controls the display device to display the image on a display target of the image, the display target including the door of the accommodation box identified by the information, and after the door has been opened, the controller excludes the door of the identified accommodation box from the display target of the image.
4. The accommodation system according to claim 1, comprising a sensor configured to detect opening of the door, wherein the sensor outputs a detection result, corresponding to opening the door, as the information on the accommodation box for which the operation regarding opening of the door has been performed.
5. The accommodation system according to claim 1, wherein the controller controls the display device to display the image on the door of the accommodation box for which the operation regarding opening of the door has been performed.
6. The accommodation system according to claim 1, wherein the controller controls the display device to display the image on a door of a predetermined accommodation box in a vicinity of the accommodation box for which the operation regarding opening of the door has been performed.
7. The accommodation system according to claim 6, wherein the controller controls the display device to display the image on a door of an accommodation box that is adjacent to the accommodation box for which the operation regarding opening of the door has been performed.

- 8.** The accommodation system according to claim 7, wherein the controller excludes, from the door on which the image is to be displayed, a door of an accommodation box that is, in an opening direction of the door adjacent to the accommodation box for which the operation regarding opening of the door has been performed.
 - 9.** The accommodation system according to claim 1, wherein the controller controls the display device to display, when the operation regarding opening has not been performed for any of the doors, an image to be displayed on a door of an accommodation box that has been predetermined, and controls the display device to display the image displayed when the operation has not been performed and the image displayed when the operation has been performed to be different from each other.
 - 10.** The accommodation system according to claim 9, wherein the controller decreases a number of the doors on which the image is to be displayed when the operation has been performed, as compared with a number of the doors on which the image is to be displayed when the operation has not been performed.
 - 11.** The accommodation system according to claim 9, comprising an interface configured to receive an input from a manager of the accommodation system, wherein the controller receives, via the interface, whether or not to perform a display process of the image when the operation regarding opening of the door has been performed.
 - 12.** The accommodation system according to claim 1, wherein the display device is a display disposed on a surface of the door.
 - 13.** The accommodation system according to claim 1, wherein the display device is a projector that projects an image on a surface of the door.
 - 14.** The accommodation system according to claim 1, wherein the image displayed on the door is an advertisement image including advertisement information.
 - 15.** The accommodation system according to claim 1, wherein the image displayed on the door is an announcement image including predetermined announcement information different from advertisement information, or an entertainment image including information that entertains a user of the accommodation box.
 - 16.** A control method for an accommodation system including a plurality of accommodation boxes each having a door and disposed so as to be arranged in an in-plane direction, and a display device configured to display an image on a door of an accommodation box of the plurality of accommodation boxes, the control method comprising: acquiring information on the accommodation box for which an operation of opening the door has been performed; setting the door, on which the image is to be displayed, in accordance with a position of the accommodation box identified by the information; and controlling the display device to display the image on the set door.
 - 17.** A non-transitory tangible storage medium storing computer executable instructions for causing a controller of an accommodation system including a plurality of accommodation boxes each having a door and disposed so as to be arranged in an in-plane direction, and a display device configured to display an image on a door of an accommodation box of the plurality of accommodation boxes, to execute a process comprising: acquiring information on the accommodation box for which an operation of opening the door has been performed; setting the door, on which the image is to be displayed, in accordance with a position of the accommodation box identified by the information; and controlling the display device to display the image on the set door.
-